KaxaNuk
Sharing knowledge

INVESTMENT BOOTCAMP

SIMULATE YOUR BOOKS AND ATTRIBUTE THEIR RETURN.



05 From your S04 books to their track record net of costs, against the KN US Equity Core and SPY, and where every point came from.

IN THIS SESSION

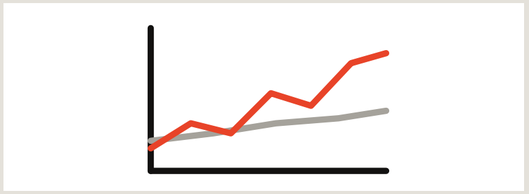
01 SIMULATE

02 COMPARE

03 ATTRIBUTE

04 WRAP-UP

FROM YOUR WEIGHTS TO WHERE IT CAME FROM.

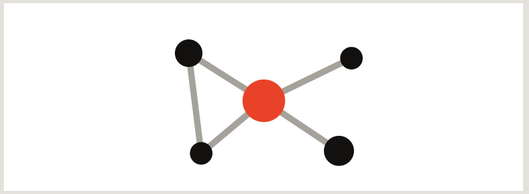


1 SIMULATE

Your track record, net of costs.

- The Backtest Engine, step by step
- Commission, slippage and reserve
- Net, or nothing

You leave with: your track record

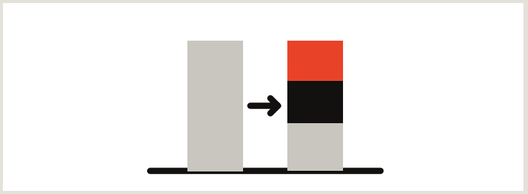


2 COMPARE

Against whom, and with which metrics.

- The KN US Equity Core and SPY
- Sharpe, Sortino, drawdown, tracking error
- Excess, active and Jensen's alpha

You leave with: your table



3 ATTRIBUTE

Where every point came from.

- The KN US Equity Factor Model
- Factors, sectors and idiosyncratic
- Two variants, component by component

You leave with: your attribution



4 WRAP-UP

What you learned, and what comes next.

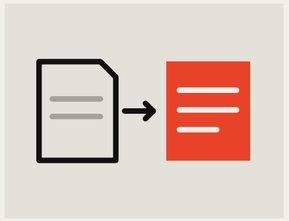
- The session's ideas, in your words
- Homework: your findings, in writing
- Next: paper trading, in S06

You leave with: your plan for S06

EIGHT STEPS. IN THIS SESSION, THE FIFTH AND SIXTH.

S02 TO S04 | THIS SESSION | S06 | BEYOND THE COURSE

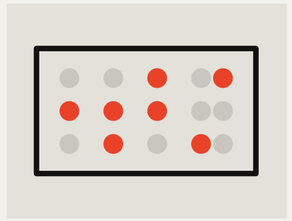
1 2 3 4 5 6 7 8



RESEARCH

S02

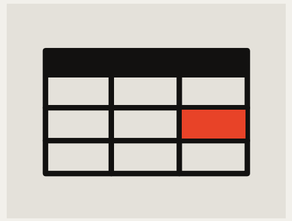
Your objective, from your library · OBJECTIVE.md



UNIVERSE

S02

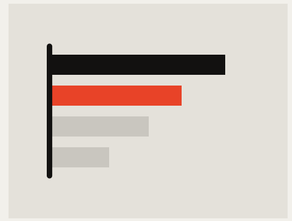
Which companies and when; delisted included · Universe/



DATA

S03

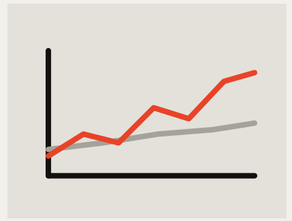
Your features, and whether they carry a signal · Data/



PORTFOLIO

S04

Your hypothesis, then the portfolios · Experiments/

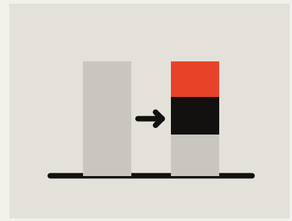


BACKTEST

S05

Your portfolio through time, net of costs

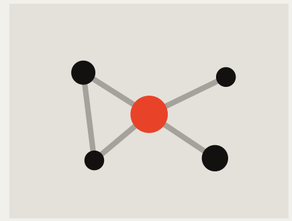
STEPS 5 AND 6



ATTRIBUTION

S05

Where the return came from



PAPER TRADING

S06

Tested without real money · Paper_Trading/



PRODUCTION

outside the course

From S04's homework

your books, your two cost rows, and N.

FROM YOUR WEIGHTS TO A TRACK RECORD.

ARYA · TREND-AND-VOLUME

Price my three books, with my two cost rows. 1

With your weights and your costs. Go ahead?

go ahead

Complete: every day, priced. 2

EXPERIMENTS/EXPERIMENT_1/BACKTEST/

```
In: portfolio_weights.csv  sums to 1
In: the prices
Out: the track record      every day
Out: Daily_Weights         the weights as held   3
Out: the metrics
```

Costs: commission, slippage, reserve.

Benchmarks: KN US Equity Core and SPY.

Metrics: CAGR, Sharpe, Sortino, drawdown, VaR.

1. Your weights and your costs, from the blueprint.
2. Check it ran complete.
3. The daily weights: attribution reads them.

`backtest-engine-runs`

HOW THE BACKTEST ENGINE SIMULATES.

RULE	WHAT IT MEANS
Decide at t – 1, trade at t	No order uses a price that did not exist yet.
Commission and slippage	Every trade pays; a commission is measured, not assumed.
A cash reserve	It pays the commission; without it, a run stops mid-window.
No price, no run	It refuses a position left without a price: sell it first.
Weights that drift	Between rebalances, each weight moves with its price.
Years of 252 days	Its CAGR counts trading days, not calendar days.

BEFORE READING IT check it priced every day you asked for: a short run looks clean too.

NET, OR NOTHING.

BY TRADED VALUE	EVERY MONTH	WHEN THE SIGNAL CHANGES
Turnover a year	242%	441%
Annual return, gross	7.0%	8.3%
Annual return, net	6.5%	7.4%
Sharpe, net	0.30	0.33

EVERY POINT OF TURNOVER COSTS 10 bp per side, an assumption; your costs, from the blueprint.

AGAINST WHOM YOU COMPARE.

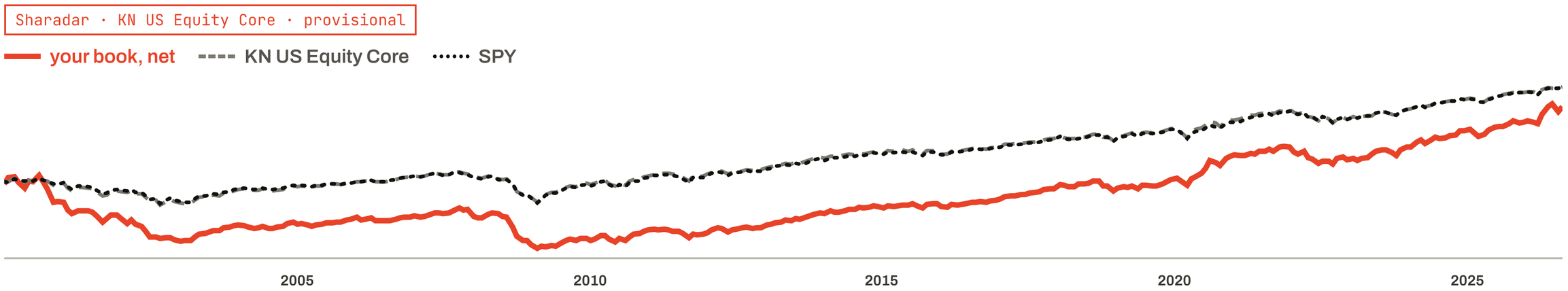
BENCHMARK	WHAT IT IS	WHAT FOR
KN US Equity Core	US large and mid caps, by full market cap, reviewed every quarter since 1998.	The same universe as your strategy: what you made over it is yours.
SPY	The S&P 500 ETF, investable today.	What anyone has with a single purchase.
Your control	Your rule without one ingredient.	First: it says which part did it.

IN ORDER first the control, then the KN US Equity Core, then SPY.

YOUR BOOK AGAINST BOTH.

GROWTH OF 1 · NET · KN US EQUITY CORE · SPY

Sharadar · KN US Equity Core · provisional



1 BECAME · YOUR BOOK	KN US EQUITY CORE	SPY
5.3	8.5	8.4

Log scale; your book by traded value, every month, net of 10 basis points per side.

Less return

and deeper falls than the benchmark: a result, not a failure.

THE METRICS, SIDE BY SIDE.

	YOUR BOOK	KN US EQUITY CORE	SPY
Annual return	6.5%	8.4%	8.4%
Volatility	25.5%	19.4%	19.3%
Sharpe	0.30	0.41	0.41
Sortino	0.42	0.58	0.58
Max drawdown	−83.2%	−54.2%	−55.2%
Beta and tracking error	1.12 · 13.5%	1 · —	—
Information ratio	−0.03	—	—

AGAINST THE KN US EQUITY CORE beta, tracking error and information ratio measure the relative.

EXCESS, ACTIVE AND JENSEN'S ALPHA.

TERM	WHAT IT MEASURES	BEFORE COSTS
Excess	Over the risk-free rate.	+216% summed
Active	Over the KN US Equity Core.	+3% summed
Jensen's alpha	The intercept of a regression against the benchmark: what is left after its beta.	–0.9% a year
From the factors	What the market, the styles and the sectors explain.	+184% summed
Idiosyncratic	What no factor explains: your signal, or luck.	+32% summed

NOT THE SAME Jensen's alpha, the intercept, discounts beta; the idiosyncratic, every factor.

THE KN US EQUITY FACTOR MODEL.

Each stock, each day

Its excess return, split into pieces that add up.

Market

The KN US Equity Core, the same for all.

Five styles

Size, value, momentum, beta, residual volatility.

Eleven KN sectors

Each stock's sector, by its SIC.

Specific

What none explains: the idiosyncratic.

Exposure at the prior close times the factor's return that day; simple returns, over FRED's T-bills.

WHERE THE RETURN CAME FROM.

COMPONENT	YOUR BOOK	KN US EQUITY CORE
Market	+218%	+218%
Styles	−40%	−0%
Sectors	+6%	−0%
Idiosyncratic	+32%	−4%
Total excess	+216%	+213%

RECONCILE FIRST the attributed total against the book's own return, before reading a line.

BRINSON-FACHLER, BY SECTOR.

EFFECT	WHAT IT MEASURES	YOUR BOOK
Allocation	Weighting more the sectors that beat the benchmark.	–19%
Selection	Picking better stocks within each sector.	+1%
Interaction	Both at once.	+21%
Active	Against the KN US Equity Core.	+3%

THE WHOLE BENCHMARK every KN US Equity Core stock, not only the ones you held.

A CLEAN ATTRIBUTION CAN STILL BE WRONG.

Before reading it

Five checks.

The day's weight

The prior close's.

The units

Simple with simple, not logs.

The benchmark

Whole, not only your stocks.

Each stock

By its identifier, not its ticker.

What it misses

An absolute cross is not relative momentum.

Each day's sums, not compounded; reconciled against the book.

TWO VARIANTS, COMPONENT BY COMPONENT.

COMPONENT	50/200	63/252	DIFFERENCE
Market	+218%	+218%	−0%
Styles	−40%	−34%	−6%
Sectors	+6%	+5%	+1%
Idiosyncratic	+32%	+52%	−20%
Total excess	+216%	+241%	−25%

THE PARTS ADD UP the total gap splits exactly across components.

SELECTION, SIZING AND TIMING.

THE QUESTION	THE COUNTERFACTUAL BOOK	WHAT IT ISOLATES
Sizing	The same stocks, same dates, at equal weight.	Whether weighting by traded value adds.
Selection	Random stocks from the eligible, at the same weights.	Whether picking those stocks adds.
Timing	The same stocks, with the dates shifted.	Whether entering and leaving when you did adds.
Before the three	Drop the positions too small to matter.	So a sliver does not drive the comparison.

ONE THING AT A TIME each counterfactual changes only one, on the same window and with the same costs.

TWO TRACKS, ONE ATTRIBUTION.

INVESTMENT LAB

NO CODE

Backtest: your book, your costs →

Attribution: the factor model →

Brinson-Fachler: the benchmark

Benchmarks: KN US Equity Core, SPY.

Factor model: KN US Equity.

Costs: your blueprint's.

Brinson-Fachler: by sector.

ARYA · TREND-AND-VOLUME

Attribute my book with the KN US Equity Factor Model. 1

Weights at the prior close, the whole benchmark, the total reconciled. Go ahead?

go ahead

Done: each component, with the section that computes it. 2

1. Code: the attribution, with its skill.
2. Read every figure from its file.

`attribution-analysis-runs`

`alpha-decomposition`

YOUR FINDINGS, IN WRITING.

ARYA · TREND-AND-VOLUME

Draft FINDINGS_1.md with the backtest and the attribution. 1

Each figure with its file. Does your hypothesis hold? You decide that.

go ahead

Drafted. Your verdict is missing. 2

FINDINGS_1.MD · EXAMPLE: THE KN US EQUITY CORE

MEASURE	VALUE	VERDICT
Net Sharpe against the control	—	
Active against the KN US Equity Core, net	-10%	
Idiosyncratic, summed	+32%	

The verdict is yours: against what your blueprint wrote before looking.

1. Ask for the draft, figure by figure.
2. The verdict, against your blueprint.

`experiment-lifecycle`

THE SESSION'S IDEAS, IN YOUR WORDS.

01

Why net, and what do you check first?

02

Excess, active, Jensen's alpha: over what?

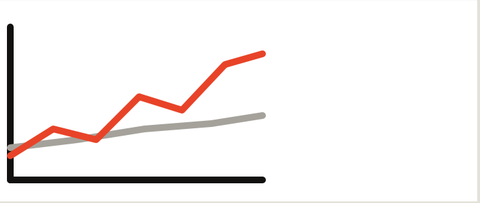
03

With which weight is each day attributed?

04

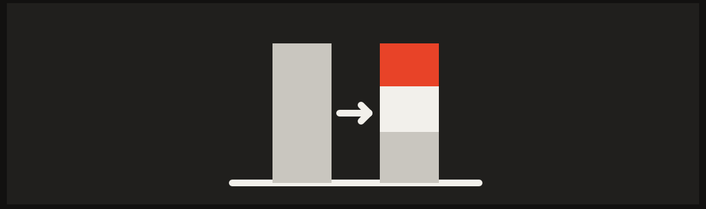
Why does the factor model barely see a cross?

BEFORE S06: SIMULATE AND EXPLAIN.



01 YOUR BACKTEST

- Your three books, net.
- Each run, complete.
- Against the KN US Equity Core and SPY.



02 YOUR ATTRIBUTION

- The factor model, reconciled.
- Brinson-Fachler, by sector.
- Your control, component by component.



03 READY FOR S06

- FINDINGS_1.md, with your verdict.
- N up to date.
- Ask `next`.
- First Steps: MF P05, MF P06, CF P06.

SIMULATED FIRST. EXPLAINED NEXT.

Your net track record, compared and explained: where every point came from.

LEARN

The Bootcamp continues, then the pillars.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

TELL US

What your attribution told you.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

Session 06 · Ready for Paper Trading? Validate, Freeze and Present

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Tools read 2026-10-13; works and links, 2026-10-11; the figures, from Sharadar prices of the KN US Equity Core and version 2 of the KN US Equity Factor Model, both still provisional, computed 2026-10-17, with 10 basis points per side. Shared under CC BY 4.0; the KaxaNuk name, logo and isotipo are not.